11 · Implementation, Monitoring & Evaluation

> **Assignment:** *"Briefly outline the plan for CDSS implementation — how will the CDSS be provided to users and maintained? Consider factors such as integration with existing systems and user training. Briefly describe an approach to monitoring and evaluation — what are the objectives at different stages of maturity? What methods will be used?"*

Part A — Implementation

11.1 Provision

Aspect	Implementation
Distribution	QR code and short link on the discharge sheet. No app store, no account, no installation
Devices	Any smartphone, tablet or desktop with a modern browser. No minimum operating system
Connectivity	Fully usable offline after the initial load. Transmission to the record takes place as soon as network coverage is available again
Hosting	Static hosting of a single HTML file — minimal infrastructure, low operating costs
Patient ↔ case assignment	Manual in the prototype. In the rollout: token-based link without password, issued at hospital discharge

11.2 Integration into Existing Systems

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WundCheck (patient)
   │  FHIR R4
   ├── QuestionnaireResponse ──▶ HIS / EPD  (daily check, all answers)
   ├── Observation ───────────▶ HIS / EPD  (traffic-light classification, as a trend
curve)
   └── Communication ─────────▶ practice inbox (only for ORANGE/RED)
```

Level	Implementation	Fallback
HIS connection	FHIR R4 **transaction bundle**, POSTed to `meta.fhir.endpoint`. Switching servers is a configuration change in the L3, not a code change	Outbound queue in the app; `Download FHIR bundle` as a manual fallback
Swiss context	Connection to the **EPD** (electronic patient dossier) via FHIR	—
Alert delivery	`Communication` resource in the same bundle, `priority: urgent` (ORANGE) / `stat` (RED), generated **only** from ORANGE upwards	E-mail notification with structured attachment
Terminology	SNOMED CT, LOINC, ICD-10	**Open:** the SNOMED CT licensing situation in Switzerland still needs to be clarified

Interoperability level achieved: structural ✅ · semantic ⚠️ partial · organisational ❌ — see 09 L4.

11.3 Training

Target group	Format	Duration	Content
Patient	Explanation during the discharge consultation + illustrated information leaflet	5 min	Scan QR code, carry out the first wound check together, explain the traffic light, name the scope limits (“when in doubt, always call")
Medical practice assistant / nursing staff	Training on the practice dashboard	30 min	Reading an alert, interpreting the trend, escalation process, acknowledgement
Medical management	Short briefing	15 min	Trend display in the HIS, limits of the system, handling discrepancies between classification and clinical findings
New staff members	Onboarding module	15 min	Because of the high staff turnover in the practice: not a one-off training session, but a fixed part of onboarding

Critical point: Patient training takes place on the day of hospital discharge — a moment of high cognitive load. Carrying out the first wound check together is therefore more important than any explanation. At the same time, this is an opportunity to check the image assignment once against the medical findings (addresses residual risk (b) from 10 QA).

11.4 Maintenance & Content Governance

Question	Answer
Who owns the clinical content?	The operating specialist department — not IT
Who is allowed to change a rule?	Change made by the clinically responsible person at L2, sign-off under the four-eyes principle
How long from guideline update to going live?	Target: **≤ 4 weeks** — L2 adaptation (1 wk) → clinical review (1 wk) → transfer to L3 + regression test (1 wk) → release (1 wk)
What happens in the event of an incident?	Incident process: report → reconstruction via the L3 version carried in the FHIR export → rule review → hotfix if required + unscheduled regression test
Who operates the technology?	Static hosting — minimal operating effort. No application server, no database on the application side

The change process in detail: 08 L3 — Versioning & Update Management.

11.5 Stakeholders

Stakeholder	Interest	Risk
Patient	Safety, orientation	Data protection concerns; feeling overwhelmed
Medical practice assistant / nursing staff	Relief for telephone triage	An additional channel = additional work if alerts are too frequent
Operating physicians	Early complication detection, quality data	Liability questions in the event of misclassification
Hospital management	Fewer readmissions, quality indicators	Investment without proven benefit
HIS vendor	—	Effort for the FHIR interface
Data protection officer	Legal compliance	Health data in the hands of patients

The critical group is the practice. If Sandra experiences the alerts as a burden, the implementation fails — regardless of the quality of the classification. Therefore: alerts only for ORANGE/RED, structured instead of free text, and measurement of the alert load as an explicit monitoring indicator.

11.6 Phase Plan

Phase	Scope	Duration	Success criterion for the next phase
0 Preparation	Clinical review of all thresholds and rules; revision of the graphics	4 weeks	Sign-off by a clinical specialist
1 Pilot	1 hospital, 1 specialty, ~30 patients	3 months	Usability demonstrated; no serious misclassification
2 Evaluation & adaptation	Analysis, rule revision, UX rework	2 months	Target values for adherence and agreement achieved
3 Expansion	Several specialties, HIS integration in productive use	6 months	Stable operation; alert load bearable for the practice
4 Scaling	Further sites, multilingualism	open	Proof of effectiveness from phase 3

Part B — Monitoring & Evaluation

We follow the maturity-stage framework of the WHO (*Monitoring and Evaluating Digital Health Interventions: A practical guide*, 2016) and the monitoring systematics from Lecture Day 6.

11.7 Monitoring Components

Component	Definition (Lecture Day 6)	Our indicators	Target value
Functionality	*Whether a system provides functions that meet stated and implied needs under specified conditions* (ISO/IEC 25000)	Loading time; error rate in rule evaluation; success rate of the FHIR export; proportion of completed runs without a technical error	> 99 % error-free runs
Stability	*Whether the system consistently works as intended — including under normal and stress conditions*	Availability of the hosting; behaviour when the network drops in the middle of data entry; data loss rate	0 data losses
Fidelity	*Whether the intervention is delivered as intended* — implementation realities may alter functionality and stability	Proportion of patients with a daily wound check; `check_streak_days`; drop-out rate and drop-out point; proportion of alerts acknowledged within 4 h	> 70 % of days with a wound check; > 90 % of alerts < 4 h
Data Quality		Completeness of the mandatory fields; rate of constraint violations; plausibility of the time series	< 5 % incomplete entries

ISO/IEC 25000 distinguishes three levels of software quality — we position ourselves as follows: *internal quality* (during development) ✅ through our desk-based testing · *external quality* (simulated environment) ⚠️ partial · *quality in use* ❌ not yet achieved.

11.8 Evaluation by Maturity Stage

Stage	M&E-Objective	Method	Instrument	Success criterion
Prototype	Does the logic work as specified?	Desk-based testing, L2 verification	Test cases T1–T19, hazard analysis	All high-risk test cases passed
Pilot	Can patients operate it correctly? Is the image assignment reliable?	Usability test; **parallel assessment** patient vs. specialist	**SUS** questionnaire; Cohen's Kappa for agreement; task completion rate; time-on-task	SUS > 70; Kappa > 0.6

Stage	M&E-Objective	Method	Instrument	Success criterion
Pilot	Is it accepted?	Semi-structured interviews, focus groups	Topic guide; **TAM/UTAUT** constructs	Intention to use positive among patients and the practice
Demonstration	Does it shorten the time to treatment?	Chart review against a historical cohort	Time from first symptom to medical contact	Reduction by ≥ 1 day
Demonstration	Does it reduce unnecessary contacts?	Analysis of practice contact data	Number of contacts without clinical consequence	Measurable reduction
Scale-up	Is it clinically effective?	**Stepped-wedge cluster design** across several wards	SSI rate; proportion of late-detected infections; readmission rate	Non-inferiority at minimum; superiority aimed for
Scale-up	Is it worth it?	Cost survey	Cost per avoided late diagnosis; cost per patient and follow-up care episode	—
Integration	Does it remain effective in routine operation?	Implementation research (**RE-AIM**)	Reach, Effectiveness, Adoption, Implementation, Maintenance	—

Why stepped-wedge and not an RCT: All wards receive the intervention, only at different points in time. This is easier to implement ethically and organisationally than a control group that permanently remains without support — and it is methodologically well suited to staggered rollouts in a care setting.

11.9 Actively Measuring Adverse Effects

An M&E plan that measures only success is incomplete. The following indicators explicitly capture the downsides:

Indicator	Why	Alarm threshold
Cases with a GREEN classification that presented to a physician within 72 h	⭐ **The single most important safety indicator of all.** It directly measures false reassurance — the risk we rate as more serious than false alarms	every case is reviewed individually
False alarm rate (ORANGE/RED without clinical consequence)	Alert fatigue in the practice; loss of trust among patients	> 40 % → rule revision
Average consultation duration in the practice	Does the system shift work instead of reducing it?	Increase → process analysis
Proportion of patients without a smartphone/internet access	Digital divide — do only the already better-off benefit?	Surveyed at every hospital discharge
Alert override rate in the practice	Are alerts routinely clicked away?	> 30 % → review alert criteria
Discontinuation of use by day (drop-off curve)	On which day do we lose patients?	—

Relation to residual risk: The first indicator directly addresses the untested points (b) and (d) from 10 QA. What we could not test before deployment must be monitored during deployment — this is the bridge between residual risk documentation and monitoring plan that Lecture Day 5 requires.

Related documents: 08 L3 · 09 L4 · 10 QA · 12 Impact